Artificial photosynthesis

Since the early 1970s, scientists have been working to develop the technology to mimic the process of photosynthesis and create liquid fuels from carbon dioxide, water, and sunlight. All three are plentiful, so if the process can be replicated it could produce an endless, relatively inexpensive supply of clean fuel and electricity.

There are two crucial steps: to develop catalysts that use solar energy to split water into oxygen and hydrogen, and to create other catalysts that convert hydrogen and carbon dioxide into an energy-dense fuel, such as liquid hydrogen, ethanol, or methanol. Scientists at Harvard University recently used catalysts to split water into oxygen and hydrogen, then fed the hydrogen, plus carbon dioxide, to bacteria. The bioengineered bacteria converted the carbon dioxide and hydrogen into liquid fuels. The next challenge is to transfer a successful lab experiment into something commercially viable.